

system requires patience and research, since – unlike a computer you have access to – you will probably not know anything about the computer and the software running on it.

Consequently, the most of the steps in hacking a remote system are those that have to do with building a repository of any and all information about the system one wishes to hack. Only once you have sharpened the knife enough, do you think about using it.

You may recall social engineering, which was covered earlier, where in order to guess a person's password, one researches as much information about them as they can. However, passwords are only one piece of the larger puzzle, and only one of the ways of bypassing security.

Even if you have the securest of passwords and the best password policies, malicious hackers will not be deterred. They will find a way into your system, often bypassing all your security by exploiting flaws in your software.

Once a hacker has enough information about the target system, he / she can begin to research the vulnerabilities in the system and use them to construct an attack. As an ethical hacker, it is important to keep note of the vulnerabilities exploited in each step of the way, and patch them.

3.1.2 Footprinting

The first step in gathering information about the target is the task of footprinting. Footprinting can be non-intrusive. Here by non-intrusive we mean that the process of footprinting does not rely on actively engaging the target. It is like a detective finding out addresses, and telephone numbers, not knocking on doors, and picking locks. Footprinting is the creation of a profile for the target containing all information about the target's level of security.

The target might be on the local network or on the internet, however in both cases, we try to determine as much information about the system(s) as we can. We will need to find out the IP address or the range of IP addresses which belong to the target, the domain names, the subnet, the services running on the target systems, the operating system and architecture of the systems, and as much more information as is possible.

As we have said before, you cannot blindly attack the system without first knowing what your vector of attack should be. Your goal should be to obtain as much information as possible that any malicious hacker would be able to obtain and use for advantage. Once we have this information, we know how exposed the systems are to harm, and can begin actually hacking the system to test its security.

For finding this information, there are a wide number of tools available, but a large amount of this information is simply publicly available. A